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Studying Child-Directed Speech

Vera Kempe

Abertay University, Dundee

Marisa Casillas

University of Chicago

Abstract

In this chapter we summarize the different questions asked by studies of child-directed speech (CDS) and the methodological approaches taken to answer them. We aim to show what the available evidence can and cannot tell us about the mechanisms that drive production and effects of CDS and its role in language transmission and suggest directions for future research that place CDS in the context of the cultural diversity of children's complex multi-modal socio-cultural environments within which language transmission takes place.

To understand the mechanisms of language evolution it is important to consider the nature of language input available to language learners. There is widespread acknowledgement that children's language input can differ considerably from the language used between adults. If language has evolved to be learnable (Christiansen & Chater, 2008) then it is important to identify the proximate and ultimate mechanisms that render it so. One essential question then concerns the drivers of the modifications that characterize child-directed speech (CDS). To date, CDS has mainly been studied within the field of language development research, often with limited consideration of the proximate mechanisms that underpin its production. This chapter aims to describe the different questions that have been asked in the study of CDS and the methodological approaches taken to answer them. We will try to illuminate what the available evidence tells us about the mechanisms and role of CDS in underpinning the transmission of language as a cultural trait. The chapter concludes by pointing to future research that can advance understanding of how CDS fits into the broader concept of multimodal child-directed communication and how language transmission is embedded in the diversity of children's wider linguistic, physical, and social environment.

To begin, some terminological clarification is in order. Although the research literature features different labels for CDS, e.g. 'motherese', 'parentese', 'babytalk', 'caregiver speech', 'child-directed communication' or 'infant-directed speech', each of them emphasizing somewhat different perspectives on the phenomenon, we will use CDS as a historically apt shorthand for the multitude of ways in which children at different ages and in different circumstances may be addressed by primary and alloparental caregivers. Despite the use of the term CDS we acknowledge that child-directed communication goes beyond spoken language and includes child-directed input in sign languages (e.g. Holzrichter & Meier, 2000) as well as non-verbal behaviors that accompany verbal communication like gesture, movement and tactile information (e.g. Brand & Shallcross, 2008; Abu-Zhaya et al., 2017).

The study of CDS started in the 1960s, when the field of child language development research was dominated by a theoretical account that viewed language input as merely a trigger of maturational processes driven by an innate, biologically determined capacity (Chomsky, 1957). Yet already in 1964, Ferguson noted cross-linguistic parallels in the existence of phonological simplifications of some words caregivers use when addressing infants and small children. He suggested that these words serve two functions: to evoke nurturant behaviors and to aid language learning (Ferguson, 1964). Because of the characteristic differences to adult-directed speech (ADS), the term CDS typically came to denote not just the language that is directed at children but a distinct way of speaking, akin to a speech register (but see Saxton [2008] and Weinstein & Baldwin [2022] for a critical analysis of the applicability of this sociolinguistic concept to CDS). The conceptualization of CDS as a distinct form of speech sparked a prolific line of research into the features of CDS, the factors that influence the propensity of caregivers to modify their language when addressing small children and the effects of these modifications on language development.

1. Comparing features of CDS and ADS

One prominent strand of research on CDS has aimed to characterize what distinguishes it from ADS. Much of this work does so by adopting the notion that CDS is a special form of speech rather than just a label for the part of child-surrounding language that happens to be directed at the child. Descriptions of the modifications taking place in CDS have been documented even though not every linguistic community has a distinct register for talking to children, a point we will return to below. Prior work in this research strand has typically obtained aggregate measures of one or several features, e.g., pitch, pitch range, vowel length, voice onset time of stop consonants, formant frequencies of vowels, mean length of utterances or type/token ratios of lexical items, to name a few, and compared them between CDS and ADS. The observed differences tend to form the basis for theory building about putative functions of CDS. In this section we discuss research that focusses on comparing CDS and ADS but does not test the learning outcomes associated with the observed differences. Conclusions about the function of CDS are drawn based on theoretical assumptions about the potential effects of CDS features, e.g., by stipulating that those features that differentiate CDS from ADS can be assumed to benefit parent-child interaction, communication, and language development. In Section 3 we discuss research that explicitly examines whether learning gains can indeed be attributed to the specific characteristics of CDS.

1.1. Comparing prosody and phonology

To date, most studies that compare CDS and ADS have focused on prosodic and phonological features. These studies typically analyze naturalistic recordings of CDS, either obtained in the home environment or in the laboratory. The ADS samples used for comparison tend to be recorded during conversations between caregivers and experimenters or, less frequently, other family members thereby mainly employing within-participant designs for the CDS-ADS comparison. Recordings typically span a time frame of minutes rather than hours and are subjected to acoustic analyses using software such as PRAAT (Boersma & Wenink, 2005), to measure parameters like mean, minimum and maximum fundamental frequency (F0), vowel duration, or pause duration. For comparisons to be valid, many studies adopt the approach of selecting CDS and ADS samples that are similar in context, content, and structure. To achieve this, caregivers may be asked to talk about the same child-centered actions or objects when addressing their child and an adult. One of the first studies of this type compared prosodic features of CDS and ADS directed at the other parent and the experimenter in mothers and fathers speaking Japanese, German, Italian, French, British or US English, and documented characteristic features like elevated pitch, exaggerated pitch range and slower speech rate in CDS (Fernald et al., 1989). Pitch range was found to be wider in maternal compared to paternal CDS, and all prosodic features of CDS were most exaggerated in US English. This study seeded a prolific line of research into comparing CDS with ADS prosody across a range of different languages (Cox et al., 2022).

The study of phonological features of CDS was inspired by Lindblom's (1990) notion of hyperarticulation that is deployed to accommodate the needs of listeners. Kuhl et al. (1997) were

the first to demonstrate increased vowel space, encompassed by the triangle of the ‘corner’ vowels /i/, /a/ and /u/ in 1st and 2nd formant space, in English, Russian and Swedish CDS. Their findings cemented the idea that CDS serves to support the learning of phonological categories based on statistical distributions of segmental cues, an idea that became known as the *hyperspeech*-hypothesis of CDS. However, attempts to generalize the *hyperspeech*-hypothesis to other phonemes and segmental contrasts have yielded inconsistent results. Some CDS-ADS comparisons of phonemically contrastive vowels situated inside the vowel triangle provided evidence for hypoarticulation and for increased variability in the instantiations of individual vowels (e.g. Cristia & Seidl, 2014). Comparisons of voice onset times by and large failed to show evidence for enhancement of the contrast between voiced and voiceless stop consonants in CDS (Baran et al., 1977; Sundberg & Lacerda, 1999; Englund, 2005; McMurray et al., 2013), and research on lexical tones in languages like Mandarin or Cantonese failed to produce evidence for enhanced tonal contrasts (Wong & Ng, 2021). More broadly, perceived speech clarity does not appear to reliably distinguish child-directed from overheard speech in naturalistic US English (Foushee et al., 2023). This lack of evidence for a generally enhanced discriminability of phonemes suggests that hyperarticulation is probably a heavily contextualized phenomenon. Its variability across phonemes and languages may be explained by trade-offs between prosodic and phonological features of CDS. For example, variability in vowel instantiations and reduced tonal contrasts may be a consequence of the exaggerated pitch range of CDS (Cristia & Seidl, 2014, Wong & Ng, 2021), and the general lengthening of voice onset times in CDS may be a by-product of its slower articulation rate (McMurray et al., 2013).

Inconsistency in findings with respect to hyperarticulation in CDS may be due to a range of variables that are likely to moderate the distribution of phonological cues in CDS. Some of these potential moderators are the age of the child, the timing of emergence of certain words in the child’s vocabulary, the lexical context of phonemes, the position of target words within utterances or how liable certain phonemes are to be modulated by prosody (e.g., sonorants or lexical tones are more likely to be affected by pitch modulations than obstruents). Researchers have tried to control some of these contextual variables through elicited CDS where caregivers are invited to talk about pre-selected toys or picture books in an attempt to control the lexical context, often at the expense of ecological validity. One quite labor-intensive solution to this problem involves scanning large corpora of CDS and ADS recorded in naturalistic settings and selecting a subset of utterances that are matched on a range of criteria. Using this method, Englund (2005) revealed that VOT differentiation may be increased in CDS compared to ADS for some contrastive pairs of voiced and unvoiced consonants but not others.

1.2. Comparing morpho-syntax, vocabulary, and pragmatics

The earliest studies of CDS typically relied on informants to document the lexical, syntactic and discourse differences between CDS and ADS, mainly by cataloging the specific words that were deemed to be unique to this register (e.g. reduplications, onomatopoeia, diminutives or

hypocoristics; see Ferguson, 1964; Oswalt, 1976). Subsequent studies compared corpora of naturalistic CDS and ADS in terms of frequency of certain words or constructions thought to make the task of extracting statistical regularities easier for the child, e.g., antonyms (Jones, 2007), diminutives (Kempe et al., 2001) or specific parts of speech (Adi-Bensaid et al., 2015).

With the emergence of digital storage solutions for standardized transcripts of speech corpora through the creation of the CHILDES-database (MacWhinney & Snow, 1985) researchers were able to share naturalistic CDS corpora more widely. At the time of writing, the CHILDES database contains approximately 436 corpora comprising over 70 million words uttered by over 16,000 monolingual and multilingual children in 48 languages. Because corpora that document naturalistic interactions between children and caregivers (as opposed to elicited narratives) contain roughly twice as much speech by caregivers than speech by children, CHILDES has become a formidable repository of CDS. However, the nature of CHILDES as a repository of caregiver-child interactions does not readily provide for an ADS comparison. As a result, comparable ADS corpora need to be obtained from other recorded adult-adult conversations, e.g., from conversations between college students or from written sources. Only when comparable ADS corpora have been sourced can comparisons to CDS from the existing corpora be conducted.

Comparisons to ADS to date have shown that CDS directed to very young children tends to differ in some general measures such as mean length of utterance, often taken to indicate syntactic complexity, type-token ratios of individual words as an measure of lexical complexity or frequency of certain speech acts as manifestations of caregiver responsiveness, sometimes all in the same study (e.g. Hayes & Ahrends, 1988). Studies comparing more specific features between CDS and ADS have shown that CDS is generally characterized by simplified structural features, for example, higher frequency of diminutives (Kempe et al., 2001) or being more associative and repetitive but less contextually diverse (Hills, 2013; Lester et al, 2022). However, to ascertain what role features that distinguish CDS from ADS may play in language development it is necessary to examine their effects on learning gains, as we will discuss in Section 3. A deeper understanding of the function of CDS also benefits from comparing it to ADS sampled from a greater variety of contexts and situations (Weinstein & Baldwin, 2022), and to other forms of speech directed at different addressees like foreigner-directed speech (e.g., Piazza et al., 2022), pet-directed speech (e.g., Burnham et al., 2002) or elderspeak (e.g., Kemper & Harden, 1999).

In contrast to the rich literature that compares prosodic and phonological features between CDS and ADS, studies devoted to a direct comparison of morpho-syntactic, lexico-semantic, pragmatic and discourse features are few and far between. We suggest several reasons for this discrepancy in the evidence base for CDS-ADS comparisons across linguistic domains: First, as mentioned above, the CHILDES database serves as a rich repository of speech by children and caregivers but has not been set up to document speech between adults. Second, studies of prosodic and phonological features can get by with relatively short recordings, often elicited in the lab, to obtain

enough utterance and phoneme tokens to enable reliable comparisons of relevant acoustic parameters. Studies of referential and structural aspects of CDS, on the other hand, require recordings that span multiple hours to extract sufficient distributional information for a reliable comparison, which renders such studies much more cost- and labor-intensive and makes it more difficult to employ within-participant study designs.

Finally, the kinds of situational contexts typically chosen to obtain CDS recordings (e.g. play time, meal time, grooming and bed-time routines) differ markedly from the situational contexts in which adults tend to interact with each other. In most studies, ADS corpora tend to differ in context and content and therefore introduce a range of situational and contextual confounds into the comparison (Weinstein & Baldwin, 2022). Even if children and adults are placed in similar situations, speech to young children tends to focus on the here and now while adults often talk about topics displaced in space and time (Foushee et al., 2023). This makes it difficult to tease apart whether potential differences in structural features and in referential content are due to the addressee or due to the topic and the situational context, suggesting that CDS is to a large extent situationally and culturally constrained. One solution to this problem of comparison, adopted from the study of prosodic and phonological properties (Liu, et al., 2009), involves letting parents observe video-recordings of other parents interacting with their children and discussing this child-centered content with another adult (e.g., Adi-Bensaid et al., 2015), thereby equating the content of both registers to some degree. Another solution is to elicit storytelling from adults directed at different addressees, a method that has shown greater redundancy in speech addressed to linguistic novices, be it children or adult language learners (Tal, et al., 2023).

2. Studying variation in CDS

Despite the initial suggestions that CDS might be a human universal (Ferguson, 1964, 1977; Fernald, 1992), there appears to be considerable variation in its prevalence and manifestation across different societies, languages, socio-demographic strata, or in response to the needs of non-typically developing children. Studying this variation can help to pinpoint conditions under which CDS can emerge as a culturally transmitted trait.

2.1. Cross-cultural variation in CDS

Ethnographic¹ accounts of early language learning and language socialization were the first to reveal that CDS as a specialized register is not apparent in all communities (Ochs, 1982; Schieffelin, 1985; Heath, 1983), challenging the notion of universality of CDS. To gain a fuller picture, recent research has focused on documenting the prevalence of CDS across different

¹ Ethnographic research is most closely associated with the field of Anthropology. It typically involves a systematic qualitative description of the study phenomenon, situated in its lived context. It aims to consider the perspectives of the individuals who relate closely to the phenomenon as well as a full array of salient, historical, institutional, and other factors affecting how the phenomenon is observed and experienced.

societies and cultural groups more systematically. Cross-cultural comparison studies typically measure either the amount and duration of time intervals during which children are being addressed directly or the number of word tokens, word types or whole utterances directed at children. Prevalence estimates require representative sampling of the language surrounding the child. One way of achieving this is through systematic observations and coding of presence or absence of CDS within pre-specified regular time intervals (e.g., Konner et al., 1977) or through time-allocation methods where all activities of target individuals, including speech production, are systematically recorded throughout the day (e.g., Cristia et al., 2019) though this method may underestimate CDS, as in Scaff et al. (2022).

Representative estimates of CDS prevalence across communities can also be obtained from video-recordings of verbal interactions between caregivers and children taken in pre-specified situations such as play time, mealtime or other routines (e.g. Shneidman & Goldin-Meadow, 2012; Vogt et al., 2015; Tamis-LeMonda et al., 2019). More recently, day-long (long-form) audio-recordings are being used to capture larger samples of ambient language without having to take *a priori* decisions about which situations or time intervals to observe or record (see Pisani et al. (2021) for a detailed review). This approach is often supported by the Language ENvironment Analysis system (LENATM) which allows researchers to obtain audio-recordings with a device that has sufficient battery capacity for 16 hours of continuous recording and can be fastened to the child's chest in a piece of clothing. The LENATM proprietary software provides automated estimates of word counts in nearby adult speech, estimated counts of vocalizations produced by the target child, and estimated counts of speaker switches between the target child and a nearby adult (i.e., instances of turn taking). Validation studies of the algorithms for deriving these automated measures, which have been trained on US English, show that the LENATM system performs fairly well across many of the languages for which it has been used in the past. However, caution should be taken in using it with languages for which there is no existing validation. A more recent open-source Voice Type Classifier (Lavechin et al., 2020; Räsänen et al., 2021) provides many of the same measures but is trained on more linguistically diverse data though, again, validation is key to reliably interpreting automated results of this sort. In general, automated approaches can provide data on a previously unreachable scale and, when carefully employed, bring about groundbreaking insights. For example, recent work using over 40,000 hours of LENATM recordings from 1001 children across 6 continents revealed that, while children's speech-like vocalization rate was predicted by input rate and normative development, rates of nearby adult speech (likely often CDS) were not predicted by any of the usual suspected factors like child age, gender, SES, multilingualism or language (Bergelson et al., 2023).

To be clear, automated analyses are not the only approach available for day-long recordings—another option is to subject recordings to acoustic, linguistic, or image-based (for video-recordings) analyses (e.g. Ko et al., 2016; Casey et al., 2022; Montag, 2020). Still, sometimes arduous, manual annotations of daylong data that go beyond what can currently be automatized

are essential for revealing how CDS and other input sources differ between the lab and real home settings (Bergelson et al., 2019) but there is rapid progress in automated solutions for transcribing and converting audio into standardized analysis formats (Liu et al., 2023).

Because of its unobtrusiveness, the day-long audio-recording methodology is increasingly being adapted to the study of CDS beyond North America and Western Europe (Henrich et al., 2010; Singh et al., 2023). For example, day-long recordings have been used to study prevalence of CDS in rural societies, including populations of subsistence farmers, farmer-foragers, and populations transitioning from these occupations into a market integrated with the global economy. Some examples include studies of children learning Tsimane' (isolate) in the lowlands of Bolivia (Scaff et al., 2022), Quechua (Quechua)-Spanish (Indo-European) bilinguals in the highlands of Bolivia (Cychosz et al., 2021), Yélî Dnye (isolate) learners on Rossel Island, Papua New Guinea (Casillas et al., 2021) and children learning Tseltal (Mayan) in Southern Mexico (Casillas et al., 2020). Comparisons of input estimates in comparably constructed cross-cultural corpora (Bunce et al., 2020) have so far revealed that, at the day-long scale, rates of speech directed to children from adults are surprisingly similar across diverse societies, however, there are clear differences in the availability of overhearable speech and CDS from other children. That said, input rates are sensitive to how researchers sample the input and what is included as directed input, with dramatic differences in estimates from one technique to another (Scaff et al., 2022; Cristia, 2020).

Without a doubt, children have vastly different experiences learning language across diverse human societies—the ethnographic studies cited at the start of this section describe several such cases. Whether and how these differences may be picked up at the day-long scale and by automated measures is a challenge for future work. Urban and rural communities differ on a number of dimensions (birth rates, parenting strategies, presence of other children, expectations about children's capabilities and roles, education and literacy, type and division of adult labor) that are difficult to disentangle with traditional psychological methods (Cristia, 2020). Work that adequately addresses cross-cultural differences will in the future likely need to include significant ethnographic and/or qualitative elements. So far, the number of non-Euro-American societies studied is still too small to draw definitive conclusions about which factors moderate the prevalence or features of CDS.

Cross-cultural and cross-linguistic differences in CDS do not preclude the existence of some similarities. Indeed, the manifestation of CDS has sometimes been found to have similar attributes across unrelated linguistic communities. For example, prosodic contours of US English CDS expressing different intentions like prohibition, approval, attention, and comfort are recognizable in South American Shuar hunter-horticulturalists (Bryant & Barrett, 2007). A careful study of K'iche' (Mayan, Guatemala) CDS revealed a distinct register for addressing children that had some features in common with US English CDS (e.g., use of diminutives), but not others (high or variable pitch; Pye, 1986). A recent meta-analysis of prosodic and phonological features of CDS

from 88 studies confirmed increased pitch, pitch range, vowel duration, vowel space and reduced speech rate in the CDS in over 30 geographically diverse language communities (Cox et al., 2022). Yet there was also considerable variation which cannot be explained entirely by methodological differences like whether CDS was observed in naturalistic vs laboratory settings or whether it was read vs. produced spontaneously. This apparent variation around a general trend calls for further study of what drives cross-cultural differences in prevalence and manifestations of CDS.

2.2. Studying variation within linguistic communities

In a landmark study, Hart and Risley (1992; 2003) combined data from ethnographic observations and transcribed video-recordings of CDS from 42 US primary caregivers, estimating that high-SES children heard, cumulatively, around 30 million more word tokens than their low-SES peers over the first few years of life. This estimate prompted widespread concerns about linguistic deprivation associated with poverty and its potential long-term consequences for language development and academic achievement. Yet when trying to replicate this finding in several low- and middle-SES communities, Sperry et al. (2019) found considerable variation in language input to children, especially when taking into account overheard speech. A recent meta-analysis concluded that while published studies collectively support the idea that CDS quantity varies by SES, the differences are of a much smaller magnitude than estimated by Hart & Risley (1992; 2003) and vanish when all child-surrounding speech is taken into account (Dailey & Bergelson, 2022). That finding is consistent with the aforementioned much larger study of over 40,000 hours of children's home language experiences, which found no effect of SES on child-surrounding adult speech quantity (Bergelson et al., 2023). It is also important to put the observed group differences in perspective: Differences in CDS prevalence between urban and rural communities across cultures tend to exceed the ones observed between high and low socio-economic strata within Western industrialized societies (Cristia, 2022).

As we have seen, for many children, CDS may not be the primary source of language input. Yet what children learn from surrounding speech not directly addressed to them is far from clear (Akhtar, 2005, Weisleder & Fernald, 2013; Foushee et al., 2023) and may depend on factors such as child age, what is being learned (e.g., words vs. facts. vs. prosodic patterns) and what attentional stance is encouraged in each culture and community. For example, while joint attention during dyadic interaction is encouraged in middle- and high-SES Western communities, other communities may expect children to learn from observing how communication is conducted by others (Lancy, 2015; Rogoff et al., 1993). Future research should prioritize the study of the different paths to language that result from variation in how child-surrounding language is distributed across speakers and communicative contexts.

More recent approaches are moving away from the focus on CDS 'quantity' as a predictor of language development by emphasizing the importance of 'quality', particularly the content of caregivers' contingent responses to scaffold child language beyond the current point in

development (Masek et al., 2021; Rowe et al., 2012). It can be argued, however, that the quantity-vs-quality debate distracts from the real goal of research in this domain: to gain a contextualized perspective on the role of children's language input (Casillas, 2023). Caregivers talk differently from family to family, not just due to differences associated with SES (Hoff-Ginsberg, 1991), but also differences in linguistic and cultural context, as well as in immediate circumstance. All these factors shape the ways in which children experience language in their social worlds.

In addition to variation in CDS across social groups there is also variation associated with different types of clinical delays or disorders relating to communication and language—sometimes arising spontaneously and sometimes arising from clinical advice that parents are given. At present, most research of this type concerns CDS directed to children diagnosed with or at high assessed risk of Autism Spectrum Disorder (ASD, primarily verbal autistic children), Down Syndrome and other learning impairments as well as children with hearing impairments (especially following receipt of hearing aids or cochlear implants). Surprisingly few studies have been conducted on speech directed to blind children (Campbell & Bergelson, 2022; Perez-Pereira & Resches, 2008) or children who stutter (Zebrowski et al., 1996). In this work, researchers typically compare CDS addressed to children diagnosed with one of these clinical conditions to CDS addressed to age- and/or language-matched peers not diagnosed with any clinical conditions relating to language or communication.

These studies, while often carried out in clinical settings with the aim to optimize language provision, are also instructive about deficit views of development—they inform us about how parental and clinical assumptions about children's capacities can shape children's language input, including CDS, and give us insight into the ways in which caregivers are able to accommodate a wider range of children's communicative behaviors. For example, when addressing verbal autistic children, caregivers have been shown to display increased responsiveness compared to interactions with non-autistic children. They have also been shown to produce more information-salient structural features such as yes/no questions, lower mean lengths of utterances, or more concrete nouns (Curcio & Paccia, 1987; Konstantareas et al., 1988; Walton & Ingersoll, 2014; Tubul-Lavy et al., 2020; Bentenuto et al., 2021; Onnis et al., 2021), which are indicative of their accommodation to the communicative needs of different children. Children with Down syndrome and other learning disabilities, on the other hand, while being exposed to more exaggerated prosody (Fidler, 2003), may encounter less child-appropriate transparent word labeling, less decontextualized language, and fewer questions — features that may reflect an underestimation of these children's cognitive capacities (e.g., deFalco et al, 2011).

Finally, studies on CDS to hearing-impaired children with cochlear implants or hearing aids show that many features are more like speech directed at experience-matched rather than age-matched controls, showing caregiver accommodation to the needs of the child. For example, when addressing hearing-impaired children caregivers exaggerate prosody more than when addressing

age-matched controls (Bergeson et al., 2006). They also may adjust syntactic complexity in line with the child's linguistic experience and fine-tune it to the child's specific lexical knowledge (Odijk & Gillis, 2021). So far, the evidence suggests that, although caregivers tend to accommodate children's hearing status, there is still considerable variation in the speech directed to hearing-impaired children (Arjmandi et al., 2022).

Overall, the study of CDS produced for children with clinical conditions including ASD, Down syndrome, visual impairment, hearing impairment, and more can yield new insights into how accommodation and responsiveness to children interact with specific beliefs and attitudes about children's capabilities. The main caveat associated with this research strand is that most studies have been conducted in Western industrialized countries. Because disability is a socially constructed category (Scheer & Groce, 1988), the parental response to it must be understood in its cultural context, making it difficult to disentangle the effects of cultural expectations about these children from the effects of caregivers' capacity for accommodating them.

In sum, much more work is needed to understand how language transmission and language development proceed under such immense variability in conditions across and within societies, both in terms of how they are supported by CDS but also by the wide variety of other linguistic and communicative behaviors children encounter and engage with in their social environments.

3. Studying language learning outcomes from CDS

Even though CDS only forms one part of the language surrounding children, it is canonically considered the part that is most important for language development, at least in some industrialized societies. Consequently, much research has focused on identifying which of its features can facilitate the aspects of language learning that are most strongly associated with social and economic success in such societies (e.g., boosting vocabulary size and literacy). Studies that examine the effect of CDS on language development can be divided into three major strands: (a) examining children's preference for, or orientation to, CDS compared to ADS, (b) testing the association between certain features of CDS and learning outcomes and (c) investigating what features of CDS aid certain aspects of language learning over and above what is learnable from ADS. Studies of the first and second strand tend to measure outcomes directly in children. Studies of the third strand require experiments, often with adult participants, since a natural control of children learning language from ADS does not exist. All three strands also use computational models to estimate learnability.

3.1. Studying infant preference for CDS

The first study to experimentally investigate the preference of pre-linguistic infants for CDS (labeled infant-directed speech, 'IDS', in this literature) used the operant auditory preference procedure where infants are rewarded with a speech signal for turning their head to one or the other

speaker mounted to each side of their seat (Fernald, 1985). Four-month-old infants turned their heads longer in the direction of CDS showing that it captured their attention more than ADS. Subsequent studies tested infants from as young as two days (Cooper & Aslin, 1990) and up until and even beyond one year of age, using a variety of procedures. While some studies presented naturalistic recordings of CDS and ADS, others controlled the distinguishing prosodic features, like elevated pitch and exaggerated pitch range. A meta-analysis of 34 studies (Dunst et al., 2012) confirmed a general preference for CDS with a medium-to-large effect size.

Infant studies are notorious for their small sample sizes which are liable to false positives. To rectify this, a cross-linguistic multi-lab consortium study aimed to replicate the CDS preference in 2,329 infants aged 3–15 months acquiring 11 different languages using the three most common procedures to assess infant attention: the single-screen central fixation procedure, the head-turn preference procedure as well as eye-tracking (ManyBabies, 2020). The findings confirmed a general infant preference for CDS, although the effect size was in the small-to-medium range, i.e., somewhat smaller than estimated by the Dunst et al. (2012) meta-analytic estimate. It also demonstrated that the CDS preference increases with age, is strongest for a familiar language and when tested with the head-turn preference procedure. These behavioral findings are complemented by infant neuro-imaging studies, using event-related potentials (e.g., Zangl & Mills, 2007) and functional near-infrared spectroscopy (e.g. Saito et al., 2007; Sulpizio et al., 2018) to uncover the patterns of brain activation that underpin the attentional processes responsible for the CDS preference.

The meta-analytic and multi-lab evidence shows that the attention-capturing effect of CDS prosody appears to be one of the most robust phenomena in child language development research. However, very few studies have explored whether other features beyond speech prosody, e.g., infant-directed action aka ‘motionese’ (Brand & Shallcross, 2008), may indicate a wider preference for infant-directed communication. A related, albeit smaller research strand is investigating infants’ ability to discriminate CDS features that signal different emotions or communicative intentions of caregivers (e.g. Mastropieri & Turkewitz, 1999). Taken together, this research sheds light on the early processing capacities that seem to accord CDS a privileged status as a source of input for language development—children like to attend to it. Further studies into CDS preference in non-industrialized societies and societies with low prevalence of CDS are needed to test the universality of this phenomenon as a pre-requisite for language transmission.

3.2. Linking features of CDS to learning outcomes

Many of the studies discussed in Section 1 interpret the features that distinguish CDS from ADS as proxies for how easy it may be to learn certain aspects of language. Here we discuss studies that investigate the effects of CDS on child language learning directly. It is fair to say that this research strand comprises the largest proportion of published articles on CDS. It is correlational in nature, linking quantity and quality of CDS features to learning outcomes. Often, aggregate measures from

CDS corpora obtained at one point in time are correlated with child learning outcomes at a later point in time (but see Slone et al. [2023] for recent work on how not only frequency but distribution of lexical items over time can affect learning).

Learning outcomes have been studied for a wide range of linguistic input domains, e.g., quantity of certain lexical items, constructions, morpho-syntactic, pragmatic and discourse features (e.g. Newport et al., 2020; Genovese et al., 2020; Davies et al., 2020), or distributional characteristics of CDS including extra-linguistic features like caregiver gestures (e.g., O'Neill et al., 2005; Gogate et al., 2015), reciprocity and turn-taking behavior (e.g. Bedrosian et al., 1988; Kondaurova et al., 2020). The range of tested language learning outcome variables is equally broad and includes, but is not limited to, frequency and nature of infant vocalizations (e.g., Henning et al., 2005), infant word segmentation ability (e.g., Saffran et al., 1996), specific aspects of prosodic (e.g., McRoberts & Best, 1997), phonological (e.g., Sim & Brechtje, 2022), morpho-syntactic (e.g., Newport et al., 1977), or lexical development (e.g. Weizman & Snow, 2001), development of broader discourse and turn-taking abilities (e.g. Bedrosian et al., 1988; Kondaurova et al., 2020) or composite measures of receptive and expressive language using various standardized tests (e.g. Wang et al., 2020). Of particular importance here are longitudinal studies that attempt to capture how the nature of associations between CDS features and learning outcomes changes with the child's age as both the aims of child-directed communication as well as children's cognitive capacities change (e.g., Rowe, 2012). Summarizing the results of this expansive line of research in detail is beyond the scope of this chapter but a common conclusion is that CDS entails simplifications in many of these input domains that can facilitate language development, although not all simplifications are facilitators of language learning (Newport et al., 2020).

A smaller but equally important set of studies employs quantitative measures or computational modeling and machine learning to test specific theories about the nature of child language learning, using various learning algorithms that vary in the degree of implementation of bottom-up (unsupervised) vs top-down (supervised) learning processes (e.g., Gaussian mixture models, neural networks, Bayesian models, decision trees). These approaches are invaluable for disentangling effects of different input features and their interactions on language learning. For example, several studies have evaluated the informativeness of various distributional patterns of word co-occurrences in CDS for learning of syntactic categories. Classification accuracy of 3-word co-occurrence patterns, i.e., word frames like *'I see it, 'I want it, 'I got it'*, instantiating non-adjacent dependencies ('AXB', where X is the target word to be classified as noun or verb) was compared with other patterns (e.g., 'XAB' or 'ABX') across several languages. The main results are that frequent *word* frames are informative for learning grammatical categories in some languages, like English, but *morpheme* frames (e.g. Russian: *ot slona* [from elephantGEN], *k slonu* [towards elephantDAT], *pod slonom* [underneath elephantINSTR] seem to be more informative cross-linguistically, especially in morphologically rich languages like Russian or Turkish (e.g. Mintz, 2003; Chemla et al., 2009; Moran et al., 2018). Other examples involve the use of

distributional learning algorithms like word embeddings to demonstrate that causal meaning can be inferred more easily from CDS without recourse to syntax (You et al., 2021) or the use of Gaussian mixture-models and Expectation Maximization to show superior learnability of vowel categories from vowels that are prosodically exaggerated compared as in CDS to vowels that are prosodically unfocussed as in ADS (Adriaans & Swingley, 2017). The latter study exemplifies the role of CDS in cross-domain facilitation of learning, e.g., when syntactic units are identified from the distribution of prosodic features, a phenomenon termed ‘prosodic bootstrapping’ (Morgan & Demuth, 2014).

To understand how cultural differences or differences in SES affect language learning trajectories, correlational studies of this kind have directly examined the effect of variation in quantitative and qualitative features of CDS on language development. For example, Casillas et al. (2020) demonstrated that rural Tseltal Mayan children—who are typically raised in a non-child-centered manner—reach early linguistic milestones like babbling, first words and first word combinations at roughly the same time as their middle-class Western counterparts. With respect to effects of CDS variation by SES, it has been shown that greater vocabulary diversity and utterance length in maternal CDS (and not SES *per se*) relates to earlier and faster vocabulary growth for US children and that adherence to language teaching practices can mediate the effect of maternal education on language development in Chinese children (Hoff, 2003; Hoff & Tian, 2005).

3.3. Comparing learning gains between CDS and ADS

Links between quantitative and qualitative features of CDS and language learning outcomes in children illuminate the mechanisms by which children process language input. They cannot, however, provide insight into whether CDS is specifically tailored to the task of language learning, as has frequently been claimed (e.g. Ferguson, 1964; Golinkoff et al., 2015; Saint-Georges et al., 2013). For that, comparisons with appropriate ADS control conditions are needed to measure potential learning gains from CDS over and above those from ADS. ADS controls are infeasible to establish in naturalistic observations as children cannot be randomly assigned to grow up under conditions where they are addressed in the same way as adults. This line of research is therefore dominated by experimental and modeling studies.

The main aim of the experimental studies is to examine whether certain features of CDS facilitate processing of some vital aspect of language learning. Using both adult and child participants, such experiments have tested questions like whether ostensive cues accompanying CDS result in more infant gaze following (Senju & Csibra, 2008), whether presence of CDS-typical lexical modifications like reduplication (e.g. *choo-choo*, *wee-wee*) or diminutivization (e.g. *kitty*, *bunny*) result in better learning of inflectional morphology (e.g. Ševa et al., 2007) or of novel words (Ota & Skarabela, 2016), or whether CDS prosody improves detection of syntactic constituents (Nelson et al., 1989), to name a few. Some of the experiments tested perception while others examined learning from limited input over the very short time frames that can be implemented in laboratory

experiments. For the most part, such experiments have established processing and learning gains associated with the CDS feature in question thereby providing proof of concept for specific learning advantages from certain CDS features. However, they cannot establish the magnitude of gains that could emerge during prolonged learning in real-life adult-child interactions.

A more direct test of learning gains from CDS is possible using quantitative estimates and computational models to compare learning from large-scale naturalistic ADS and CDS corpora. These studies, which emerged within the last decade, are starting to paint a more complex and nuanced picture of what specific kinds of learning benefits CDS can have, over and above ADS, in each cultural context. For example, discriminability of phonemes in CDS is enhanced in some languages, e.g., in English CDS (Eaves et al., 2016) and reduced in others, e.g., in Japanese CDS (Martin et al., 2015). Crucially, the computational approach makes it possible to examine the contribution of various types of information like acoustic, phonological or phonotactic cues to different aspects of learning like word-form learning (Guevara-Rukoz et al., 2018) or word segmentation (Ludusan et al., 2021; 2022). It also makes it possible to investigate the suitability of different learning algorithms to capture this information (Ludusan et al., 2016, 2022). Computational studies permit not just quantification of the magnitude of the CDS benefit but can also assess how much could be learned about that same aspect of language based on ADS. For example, Kodner (2019) showed considerable overlap between CDS and various genres of ADS in the prevalence of some structural properties of verbs (e.g., irregular verbs, direct-object/to-dative alternator verbs) among the most frequent verb types, suggesting that CDS and ADS may share many high type-frequency lexical items. As a result, a learning algorithm based on the Tolerance Principle (Yang, 2016), implemented to simulate English past tense *-ed* learning, achieved similar outcomes with CDS and ADS. Similarly, Ludusan et al. (2016) showed that while prosodic boundaries were easier to detect from CDS input they could also be learned from ADS input, albeit with slightly reduced outcomes, a finding that echoes results from laboratory experiments where learning of inflectional morphology was supported by diminutivization but not impossible without (Kempe & Brooks, 2005). The latter point is important for gaining a better understanding of how language transmission can proceed despite relatively low prevalence of a distinct CDS register in some communities.

4. Explaining the drivers and functions of CDS

What drives the observed differences between CDS and ADS? Ferguson's (1964) original proposal that CDS serves to aid language acquisition has had considerable influence, so much so that in the field of language development research, a facilitative function of CDS is often taken for granted. Yet surprisingly little work has addressed the proximate drivers of CDS. If CDS serves to aid language development, should we then consider it a form of teaching? If so, does this imply awareness of, and intent to teach the child how to speak on the part of the caregiver? It has indeed been suggested that CDS may be consistent with pedagogical input sampling (Shafto & Goodman, 2008) potentially rendering CDS as a tool for teaching some aspects of language (Eaves et al.,

2016). Are there mechanisms by which a teaching function may manifest without the intent to teach? Recent theorizing on the evolutionary origins of teaching behavior in general may provide some answers: Brandl et al. (2023) have suggested that the human capacity for teaching is a cognitive gadget – a culturally transmitted behavior that utilizes existing domain-general mechanisms in the service of an outcome that is adaptive under certain ecological, cultural, and social conditions. It seems reasonable to apply the same logic to CDS and to view it also as a culturally transmitted cognitive gadget.

As a cognitive gadget, CDS may capitalize on two domain-general mechanisms: affect expression and speech/communication accommodation. Specifically, the prosodic features of CDS may be expressions of the positive affect elicited by infants in their caregivers (e.g. Singh et al, 2002). Indeed, neuroimaging data demonstrate that, for mothers, exposure to their own infants is one of the most powerful positive mood inducers (Nitschke et al., 2004). In addition, caregivers may use the signaling function of exaggerated prosody to try to capture children's attention, which, in turn, may facilitate joint attention between child and caregiver, making each person's intended meanings more apparent to the other. When the attention-capturing prosody of CDS is attenuated, as it happens in depressed mothers, infant learning appears to proceed differently (Kaplan et al., 1999). Not only does CDS capture children's attention, but it can also regulate their affect and arousal, as evidenced by infants' ability to distinguish different intonation contours (e.g. Papousek et al., 1990). Furthermore, many features of CDS arise in response to children's vocal behavior (Elmlinger et al., 2019) as rewards for the child but also as a form of speech accommodation on the part of the caregiver, like when raising one's pitch during interaction with an interlocutor who is emitting high-pitched vocalizations (Ko et al., 2016). Even though there is debate regarding whether alignment and speech accommodation in general are based on automatic processes like priming (Pickering & Garrod, 2004) or involve effortful perspective taking (Epley et al., 2004), the cognitive gadget perspective on CDS would suggest that language evolution may not just be shaped by human cognitive capacity constraints (Kirby et al., 2008) but also by the effects of capabilities like affect expression and Theory of Mind on language transmission.

Conceptualizing CDS as a cognitive gadget that builds on the fundamental human capacities of affect expression and speech/communication accommodation does not preclude its use as an intentional teaching device under some circumstances. It is, however, unlikely that intentional teaching through CDS constitutes a human universal nor that language evolution is dependent on it. Instead, just like teaching of other cultural traits, intentional use of CDS to teach language may arise under conditions in which the lifetime and population productivity gain from being taught exceeds the productivity loss associated with teaching (Gurven et al., 2020). This may well be the case in many industrialized literacy-intensive societies where linguistic prowess contributes to individuals' lifetime productivity. Future research, including modeling work, will have to try to map out under what conditions cultural transmission processes may have combined, consolidated,

and directed these domain-general capacities towards aiding the development of language and language-related skills (e.g., literacy).

A cultural evolution-perspective on CDS as a culturally transmitted cognitive gadget can also help us gain a deeper understanding of how variation of CDS across cultures and social strata interacts with humans' highly robust ability to learn language: For millennia, many thousands of languages have been transmitted across generations in extraordinarily diverse environments. The evidence to date makes it clear that there is not just one way in which language can be learned, nor one set of functions that it can serve: language allows humans to engage in many different types of encultured behaviors. Yet in most industrialized and developing countries research emphasis is often placed on vocabulary development as a prerequisite for literacy and STEM education. This emphasis elevates lexically diverse, standardized, child-centered speech to a gold-standard for language input and perpetuates societal expectations about the nature of “optimal” CDS—expectations which are culturally, societally, and institutionally transmitted. A by-product of such standardization efforts is the marginalization of under-powered and under-represented linguistic communities and their language transmission practices. Concomitantly, many efforts to boost children's early vocabulary development through interventions lack an awareness of, and engagement with the broader societal pressures that strongly reinforce the stark inequalities in socio-economic conditions they are attempting to overcome (see Rowe & Weisleder, 2020 for a more in-depth review and Loewenstein & Chater [2022] for a broader perspective on the inadequacy of individual psychological solutions to societal problems).

5. Future Directions

Over the past six decades, the emphasis of research has shifted from a characterization of CDS as the main source of child language input to a more nuanced understanding of CDS as just one of many different aspects of child-surrounding language and communication. Trying to capture how the richness of interactions with children enables development of language and other cognitive and social capacities is increasingly informed by broader theoretical perspectives put forth by bioecology (e.g. Bronfenbrenner, 1979), nonlinear dynamics system theory (Thelen & Smith, 1994), evolutionary psychology (Csibra & Gergely, 2009) and cultural evolution research (Creanza et al., 2017, Brandl et al., 2023). These perspectives acknowledge the multi-systemic and multi-functional nature of communication that children are involved in and provide avenues for contextualizing the diversity of paths to language that children may take and, consequently, the different conditions under which language transmission proceeds. Specifically, we suggest that future research on child-directed and child-surrounding communication should consider the following research directions:

1. A first move will be to shift away from viewing child-directed language merely as a source of data from which children extract statistical and referential information, and to instead consider it part of a broader repertoire of culturally transmitted child-rearing behaviors and

socialization activities. Doing so will provide deeper and more nuanced insights into how different proximate mechanisms (i.e., caregiver affective signaling, child affect regulation, speech/communication accommodation or intentional teaching) interact to generate the language children are exposed to. Currently, our understanding of the different input types and different facets of learning is still limited. The cross-cultural work conducted to date is telling us that we are missing crucial information about different types of communication that includes the child, including talk directed to other (sometimes older) children and talk in which the child is a secondary addressee. Such a shift in emphasis can help to uncover the processes that mediate how ecology and culture shape the communication that children partake in. A cultural-evolution perspective may prove especially useful here: we can think of the different strategies employed in child-directed communication as culturally transmitted traits that encompass conventionalized templates for how to conduct interactions with children. Such templates can then be modified flexibly according to the needs of the interlocutors and the constraints of the situational context. This view shifts the explanatory burden of how child-directed communication supports language transmission away from constraints acting on the individual towards communicative strategies that are themselves transmitted as adaptive behaviors in various cultural and ecological contexts. This, in turn, will offer a broader perspective on the multiple constraints that may act on language evolution.

2. The nature of communication that is directed at the child, surrounds the child or is relevant to the child more broadly needs to be viewed in the context of the richness of human activities and interactions in general. To date, a disproportionate emphasis on play with toys and picture-book reading in laboratory studies and on dyadic mother-child interactions may have provided a distorted picture that misses the diversity of communicative situations in which infants and young children acquire language. Moreover, comparison of child-directed to adult-directed communication is often restricted to a few standardized adult interactions (e.g., interviews between mothers and experimenters) which are of limited ecological validity and do not even come close to capturing the broad variety of adult communication (Weinstein & Baldwin, 2022). In addition, the existing comparisons between CDS and ADS tend to focus on a few domains of language, mainly prosody and phonology, at the expense of others, including morpho-syntactic, lexico-semantic, and pragmatic features. Considering a broader range of linguistic features, social situations and interlocutors will provide a better understanding of what is truly unique about interactions with children and what is drawn from the broad and flexible repertoire of general human communication strategies. Of specific importance will be a move away from aggregate measures of CDS features towards a greater emphasis on how these features change dynamically across different social and physical activities (Casillas, 2023).

3. A deeper appreciation of the multi-modal nature of interactions that are relevant to the child will provide a better understanding of developmental cascades that support language development in the context of the development of other capacities. Expanding the focus on other modalities (e.g., gesture, gaze, touch) will shed light on how the different proximate functions of child-directed communication manifest themselves across different interaction channels. For example, in some cultures and situations, caregivers' affect signaling and regulation of the child's affective states may be achieved through tactile rather than verbal interaction (e.g., de León, 2021; Abu-Zhaya et al., 2017). The same may hold for how joint attention is initiated – there may be differences with respect to whether the primary channel is verbal, gestural, or tactile (Abels, 2020). Moreover, depending on culture, intentional teaching efforts that involve scaffolding and modifying features of the environment may be directed towards skills other than language development (e.g., motor development, kinship knowledge; Karasik et al., 2015; Casillas, 2023). Capturing the interplay and timing of interactions across modalities, domains and skills will provide a fuller picture through which to appraise the role and relative importance of child-directed communication, and language transmission, in human development. This will entail supplementing data-gathering methodologies focusing on the auditory modality with those that include information on the role of visual, tactile, and kinesthetic interactions as components of the entire embodied relationship between caregivers and children (Shai & Belsky, 2017).
4. The broadening of research perspectives advocated above necessitates a broadening of our methodological repertoire and integration of observational and experimental methods (Cristia et al., 2023). This will entail expanding and augmenting current analysis methods of long-form recordings to *in vivo* approaches that capture contextual information (e.g., including logging of situational context alongside transcription of linguistic content) but also broadening elicitation methods used in the laboratory to include diverse situations and interlocutors. It will also require adapting experimental procedures in ways that ensure their relevance to specific cultural contexts. Crucially, to expand the cross-cultural evidence base we need to continue along the path of consortium-based developmental science such as that pioneered by the ManyBabies projects (e.g., ManyBabies Consortium, 2020), but in a way that includes local researchers and under-represented communities in the co-design of studies to adequately examine the wide variety of children's paths into language.

Broadening perspectives and research approaches to view child-directed communication as embedded in diverse social contexts has the potential to offer a deeper understanding not just of the many paths to language development but also of how cultural transmission of language, and ultimately, its evolution, are shaped by the range of conditions that characterize human social diversity.

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